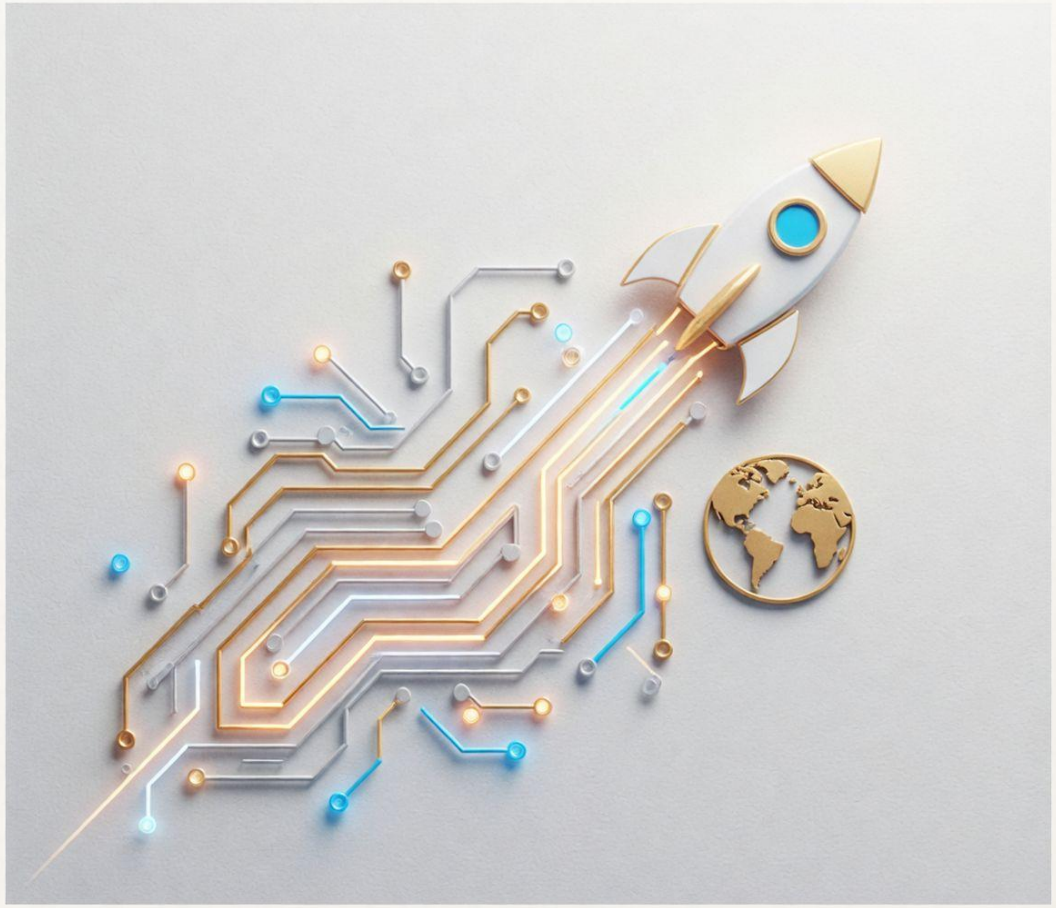
Introduction to Entrepreneurship and Sustainable Business

Ir Prof Alan Lam, JP



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Introduction to Course Leader:

Ir Prof Alan Lam, JP, PhD, SMIEEE, MHKIE
Adjunct Professor of CUHK, CityUHK and PolyU
Chairman of Sengital Group
Web: https://sengital.com and https://poffices.ai



Adjunct Professor of CUHK MAE (starting teaching in 2017):
- Course: MAEG3080 Fundamental of Machine Intelligence
- Course: MAEG5715 Artificial Internet of Things Technology
- Course: MAEG5735 Applied Computational Intelligence
- Course: ROSE5780: Special Topics in Robotics
- Course: ROSE5790: Robotics Entrepreneurship and Commercialization
- Course: GESC1250 Startup Exploration

Adjunct Professor of CityU EE (starting teaching in 2019):
- Course: EE5437 IoT Technologies for Future City Applications

Adjunct Professor of CityU BME (starting teaching in 2025):
- Course: BME6145 Applied Artificial Intelligence for Biomedical and Healthcare Applications
- Course: GenAI Technologies for Biomedical and Healthcare Applications

Adjunct Professor of PolyU EEE (starting teaching in 2025)
- Course: EEE532 Artificial Internet of Things

Adjunct Professor of PolyU MM
Adjunct Professor of CityUHK Academy
Adjunct Professor of CUHK Medicine
Advisor of KTEO of PolyU
- Training Workshops: Business Plan and Pitching
- Professional Lecturer of RTTP programme

Senior Consultant and Lecturer of GBA Business School (2022-2024)
- Course Leader: Entrepreneurial Financing
- Professional Trainer: Entrepreneurial Project

Senior Consultant and Lecturer of Left-Right Circle Academy (2024-Now)
- Course Leader: Entrepreneurial Financing
- Professional Trainer: Entrepreneurial Project

Trainer and Lecturer of HKSTP Startup Lecture (2022-now)
External Professional Consultants, University of Macau (2026-now)

Introduction to the Course

Objective:
The course “Commercialization and Entrepreneurship” aims to equip students with the essential knowledge and skills to transform innovative robotics concepts into successful business ventures. The students will engage in hands-on learning experiences, where they will explore the integration of technological advancements in entrepreneurial principles. The course is designed to foster an understanding of market needs, the development of viable business models, and the effective strategy needed to launch and grow a robotics company in today’s competitive landscape.

Justifications
As technology rapidly advances, the demand for innovative solutions in various industries increases correspondingly. Entrepreneurs with a background in entrepreneurship can effectively bridge the gap between engineering advancements and market needs. This course addresses the critical need for professionals who can navigate the complexities of launching technology-driven ventures, particularly in entrepreneurship. By focusing on practical applications and commercialization strategies, students will be better prepared to seize entrepreneurial opportunities within this dynamic field.

Course Syllabus:

LN01 Introduction to Entrepreneurship and Business
LN02 Identification of Market Potential
LN03 Product Commercialization
LN04 Company VMV and SDG
LN05 Team Formation and Value Proposition
LN06 Project Planning and Management
LN07 Business Model
LN08 Marketing and Sales Strategies
LN09 Accounting and Finance
LN10 Budget and Fundraising
LN11 Business Plan
LN12 Pitch Deck

Final Pitching Presentation

Introduction to the Course

Learning Outcomes

Upon completion of the course, students will be able to:

1. Understand foundational concepts in entrepreneurship.
2. Assess and analyze market opportunities
3. Develop and articulate a robust business model
4. Formulate marketing and sales strategies
5. Create a comprehensive business plan that incorporates financial projections, budgeting, and fundraising strategies.
6. Construct an effective investment deck to attract potential investors.
7. Deliver a compelling elevator pitch to succinctly communicate the essence of their business idea.
8. Collaborate effectively in teams to foster innovation and creativity in business development.

Deliverables will include a range of both written assignments and presentations, ensuring that students can demonstrate their understanding and application of the course content in real-world scenarios.

Assessment Scheme:

The assessment for this course will be comprised of various components, which together reflect the learning outcomes and practical applications of the concepts covered in the lectures. The breakdown is as follows:

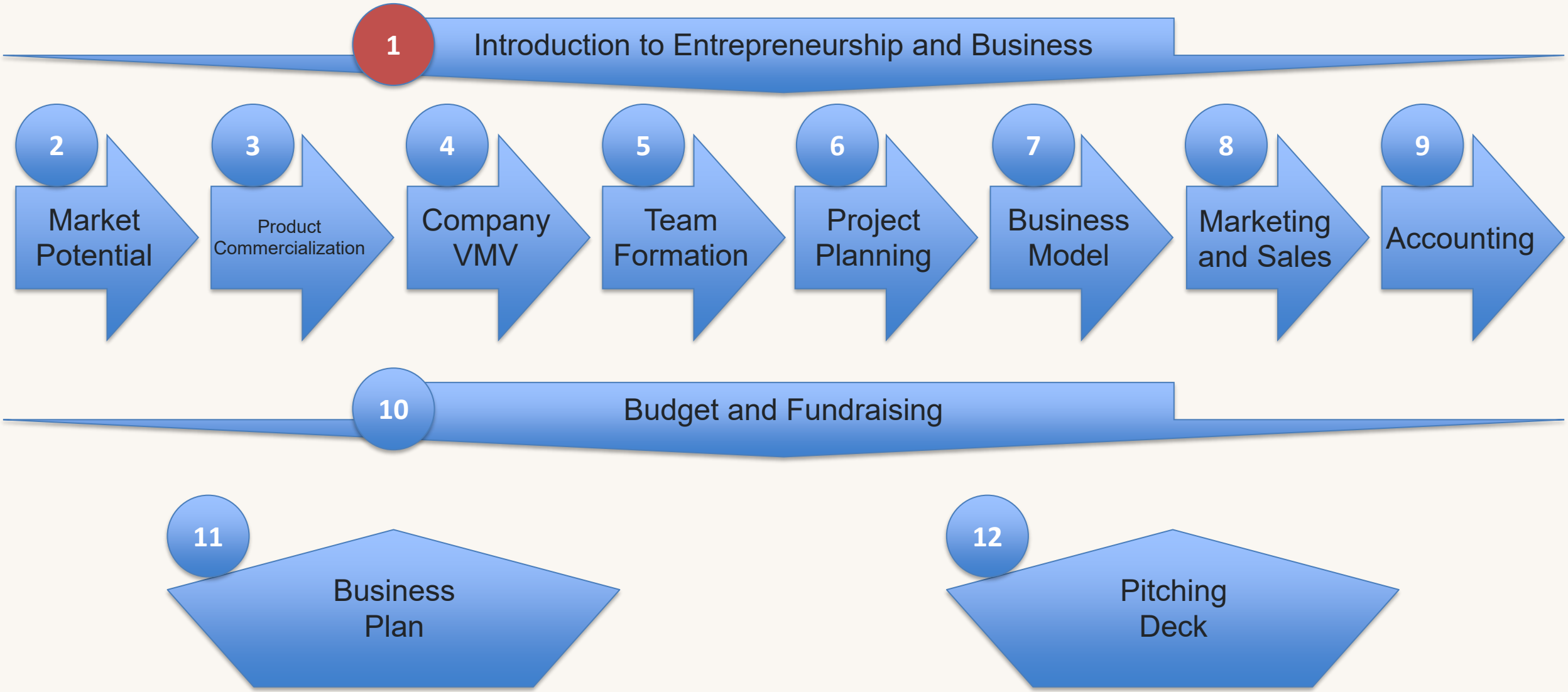
1. Proposal: 10%
2. Development Plan: 20%
3. Business Plan: 30%
4. Investment Deck: 20%
5. Final Presentation: 20%

Students will be encouraged to engage in collaborative projects that reflect innovative thinking and entrepreneurial spirit. Peer feedback will be an essential part of the learning process, allowing students to refine their ideas in an interactive environment. The course will culminate in final presentations where students will pitch their business plans to a panel, simulating real-world investment scenarios.

Introduction to the Course

Teaching Week	Date	Activity	Topic
1	11/9/2026	Lecture	Introduction to Entrepreneurship and Sustainable Business - SDGs and ESG
2	18/9/2026	Lecture	Market Potential Identification
3	25/9/2026	Lecture	Product Commercialization **Project proposal submission**
4	2/10/2026	Lecture	- Introduction to Company VMV and Lean Startup
	9/10/2026	CUHK E-Day Morning Session	- (Optional) Opening ceremony and keynote speech Visit different booths and see different demos from real startups to learn final outcomes of this course.
5	9/10/2026	Lecture	Team Formation and Value Proposition
6	16/10/2026	Lecture	Product Development
7	23/10/2026	Lecture	Business Model
8	30/10/2026	Lecture	Marketing and Sales **Development Plan submission**
9	06/11/2026	Lecture	Account and Finance - Accounting Guest Lecture
10	13/11/2026	Lecture	- Startup Budget and Fundraising
11	20/11/2026	Lecture	Business Plan
12	27/11/2026	Lecture	Startup Pitch Deck **Business Plan submission**
13	4/12/2026	Presentation	Presentation - **PPT submission**

Outline of the Course



Chapter 1: Introduction to Entrepreneurship

Ir Prof Alan Lam, JP



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Entrepreneurship Definitions and Concepts



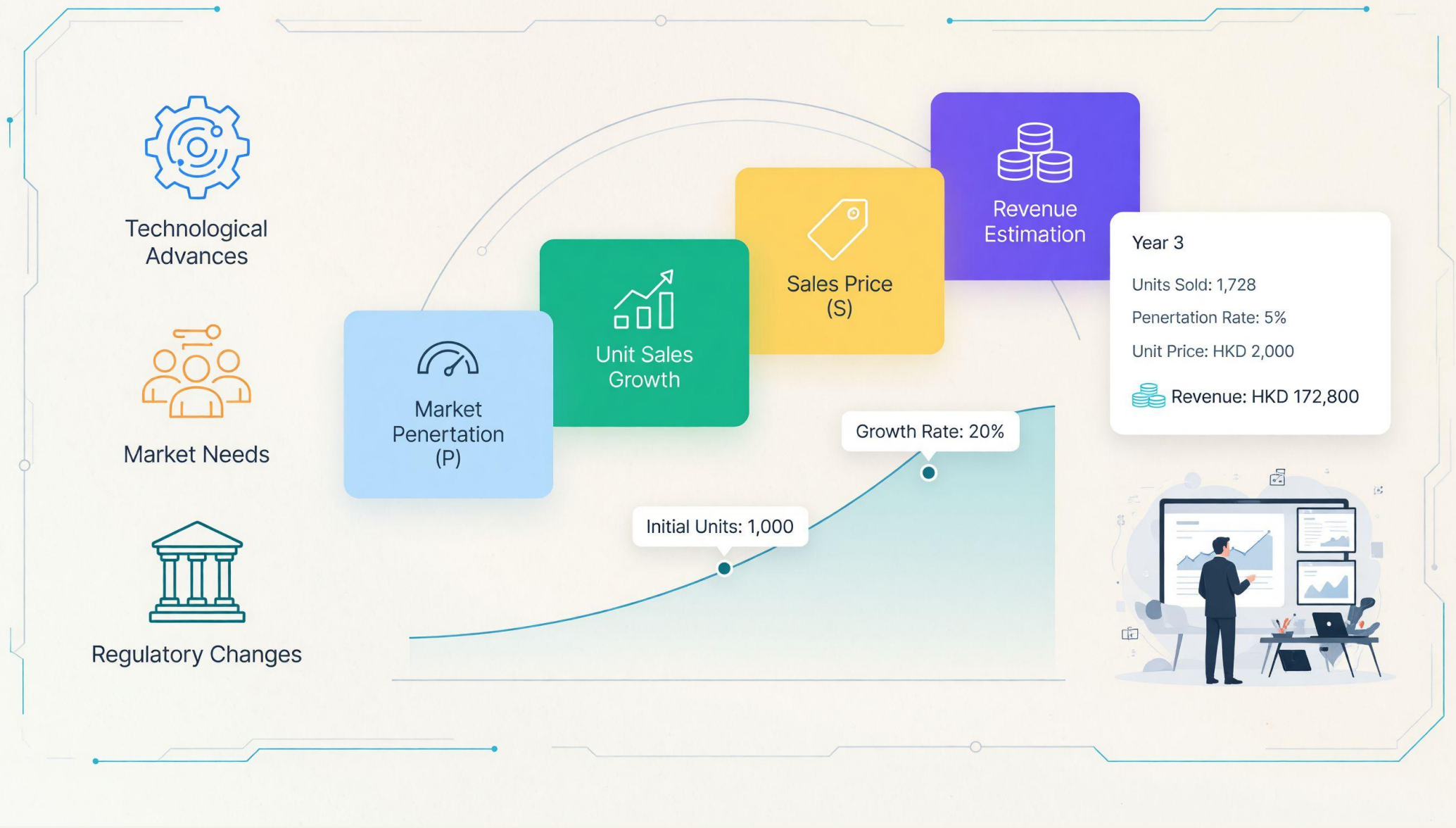
Types of Entrepreneurship



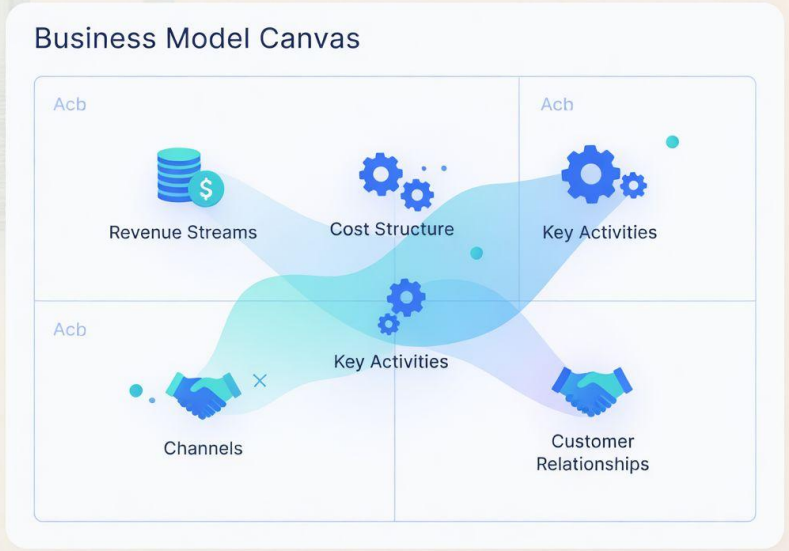
Entrepreneurship Definitions and Concepts

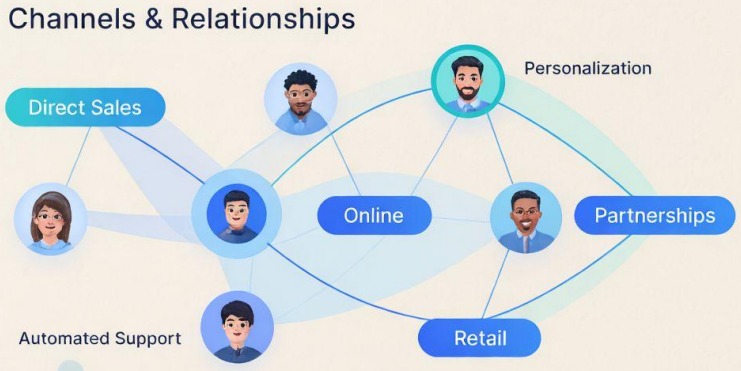


Opportunity Recognition in Electrical and Electronic Engineering

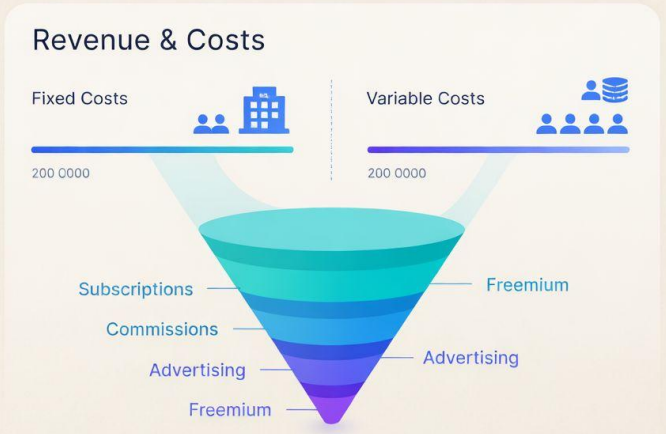


Business Models and Value Proposition







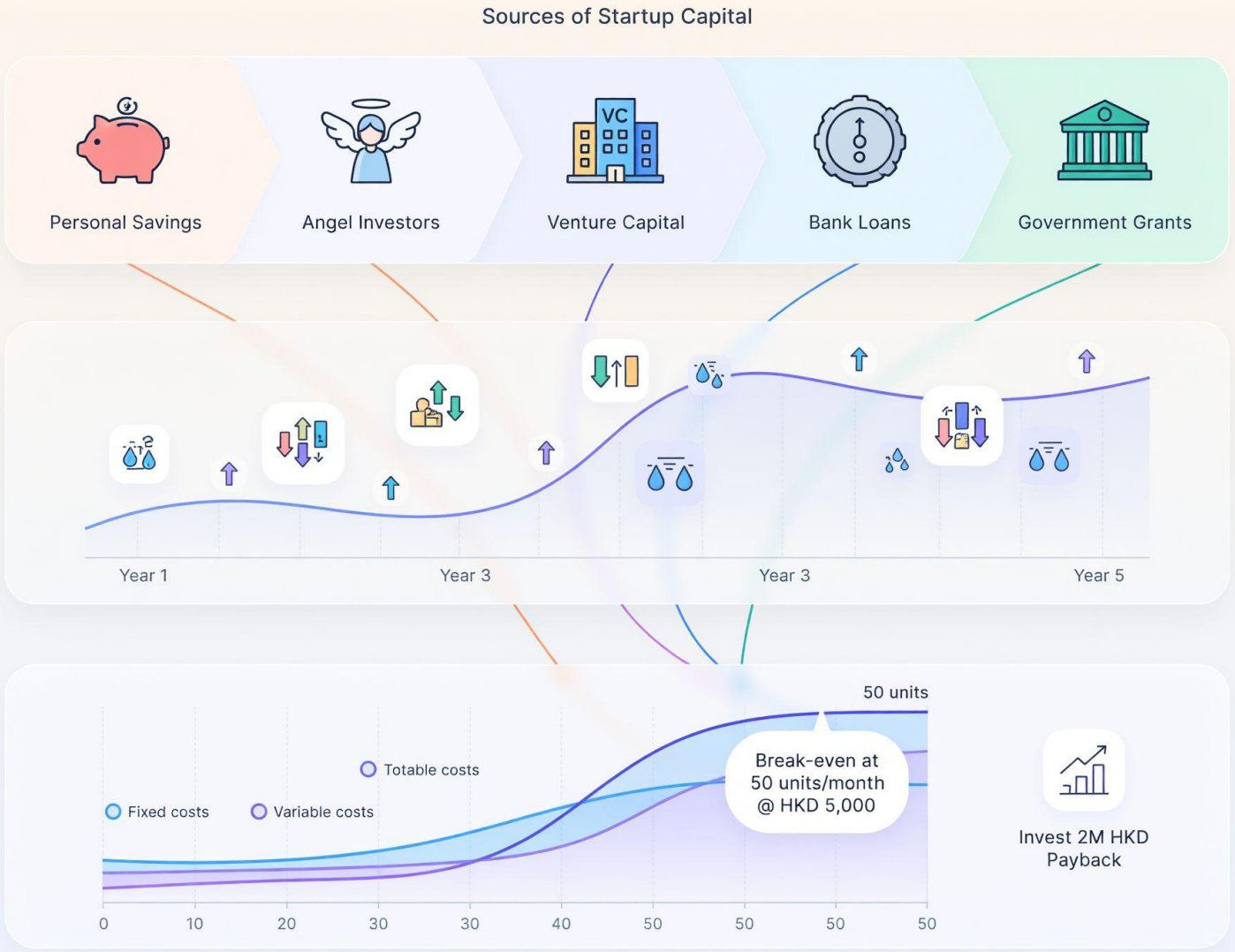




Basics of Commercialisation Processes



Funding Sources and Financial Planning



Case Studies in Electrical and Electronic Entrepreneurship



Innovation and Market Awareness

Regulatory Adaptation



Competitive Flexibility



Entrepreneurial Success Model



Key Insights for Electrical/Electronic Ventures

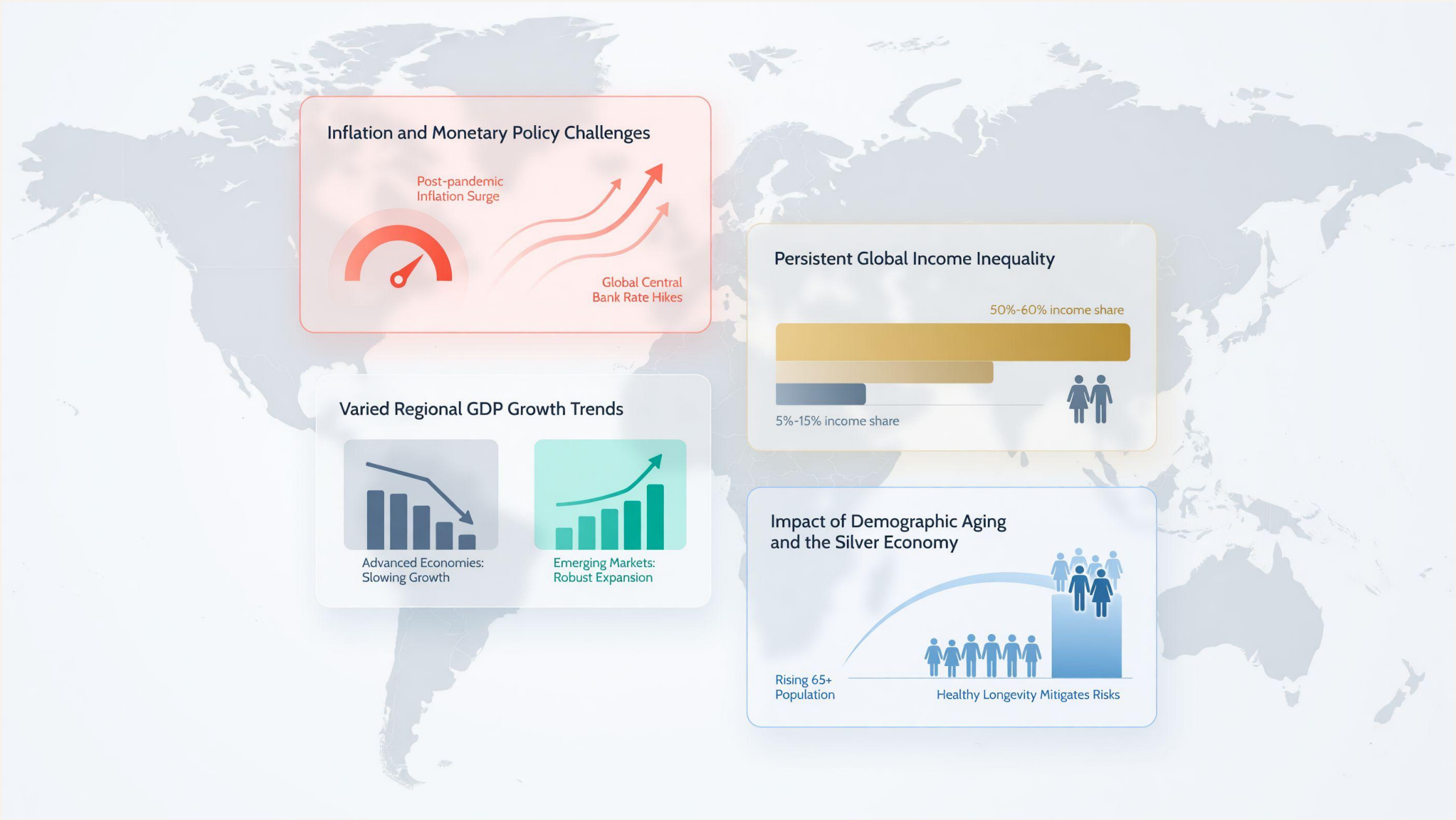
Chapter 2: Trends of Global Economy

Ir Prof Alan Lam, JP



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Overview of Global Economic Indicators



Current Macroeconomic Indicators Analysis



Trade and Global Supply Chains Analysis



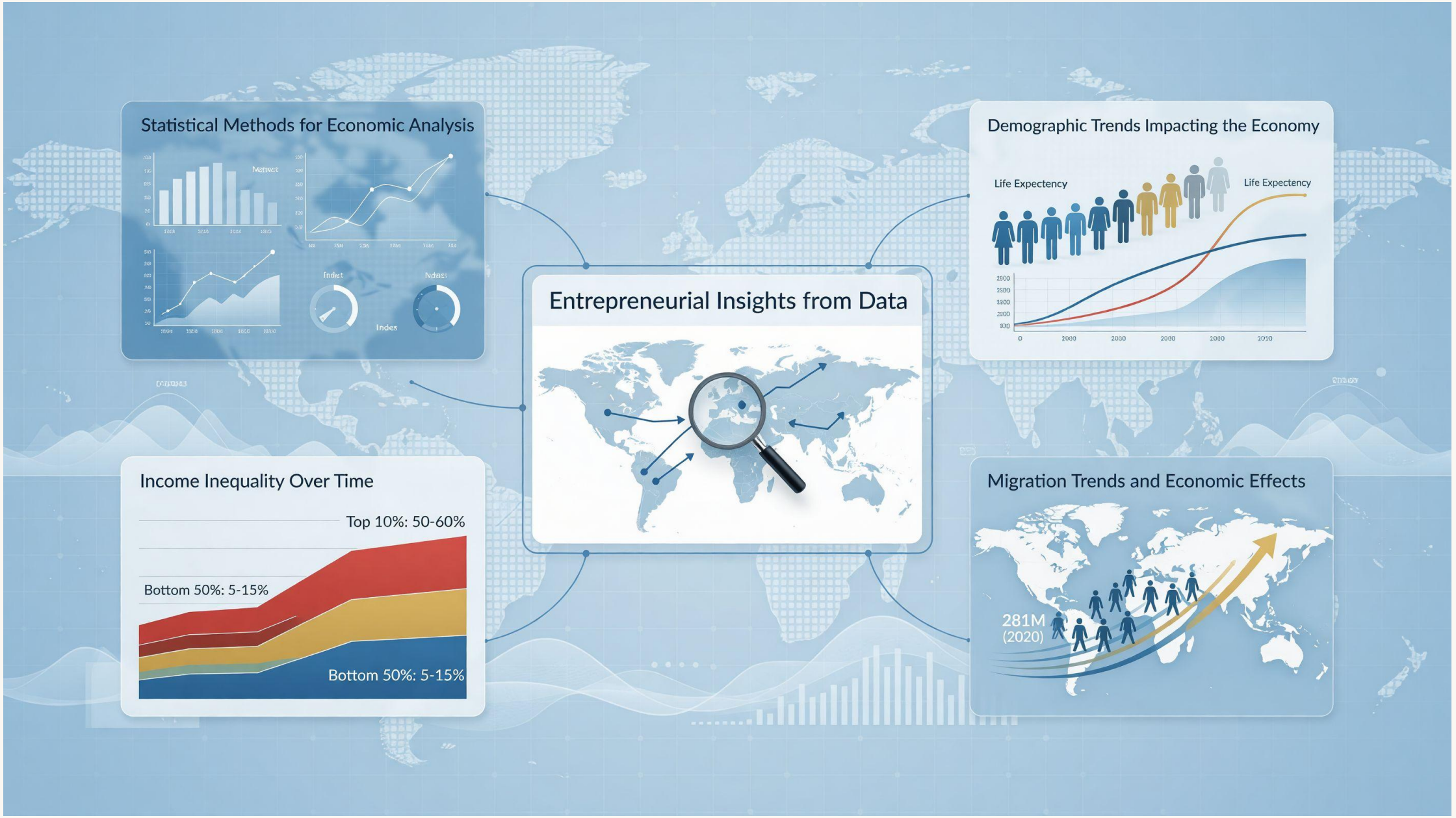
Impact of Technology on Economy



Emerging Markets and Investment Flows



Statistical Analysis of Economic Trends



Global Supply Chain Dynamics



Economic Policy and Startup Implications



Chapter 3: SDG and ESG

Ir Prof Alan Lam, JP



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Sustainable Development Goals Overview



Integration of SDGs in Business



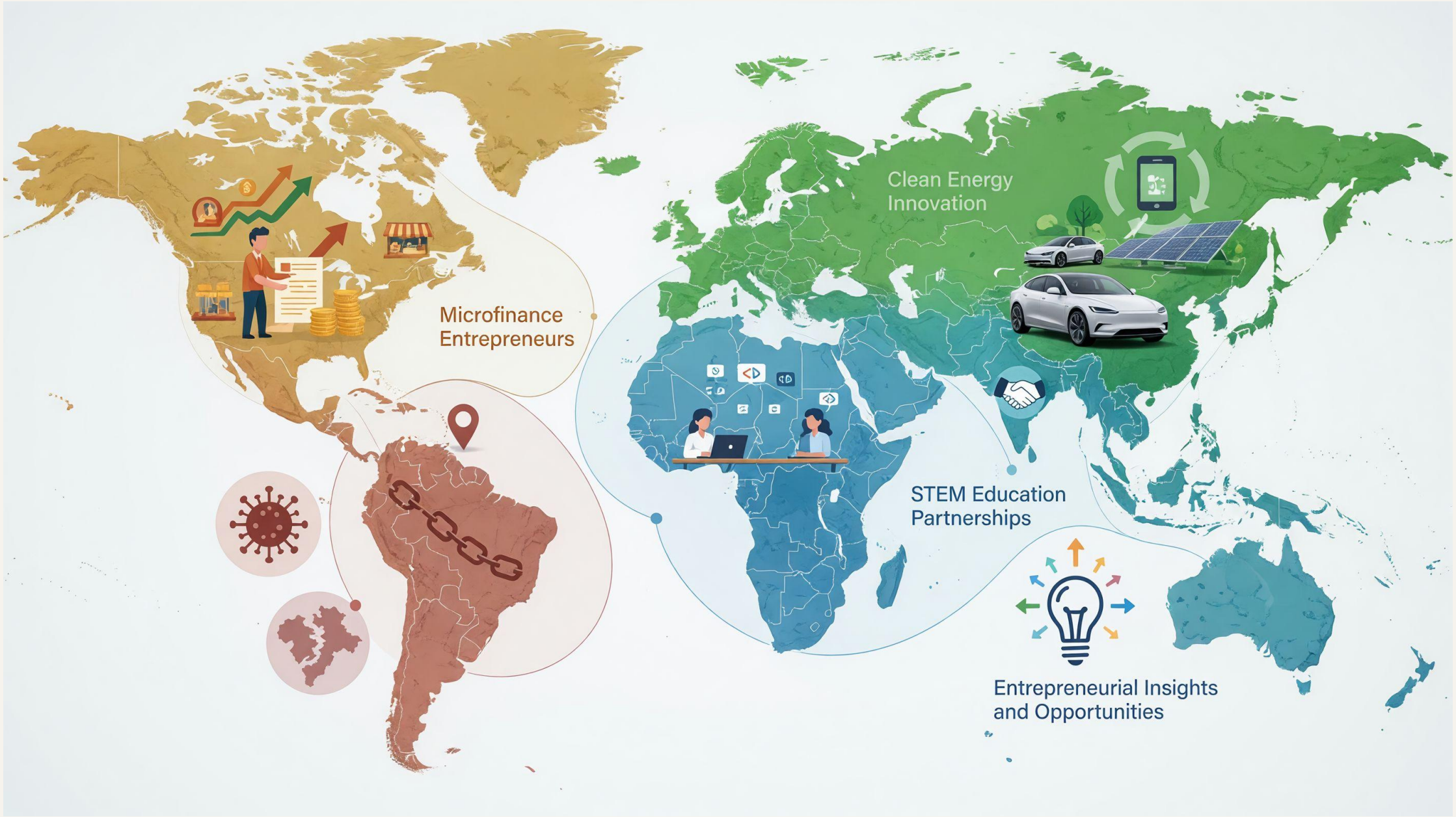
Corporate SDG Integration







Case Studies on SDG Implementation



Environmental, Social, and Governance (ESG) Criteria



Integration of SDGs in Business



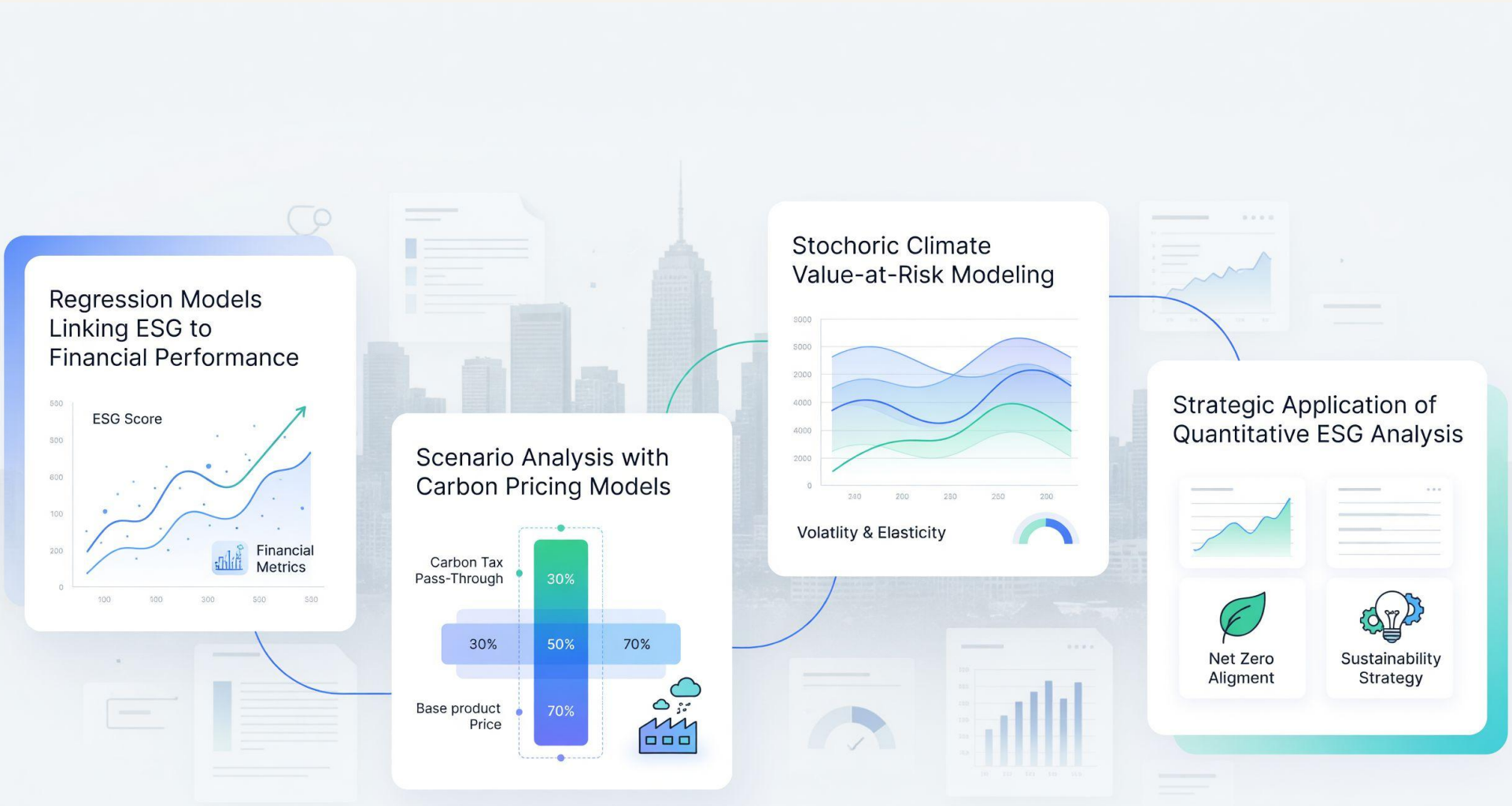
Corporate SDG Integration







Quantitative Analysis of ESG Impact



Reporting Standards and Frameworks



Step-by-step ESG Rating Calculation Example



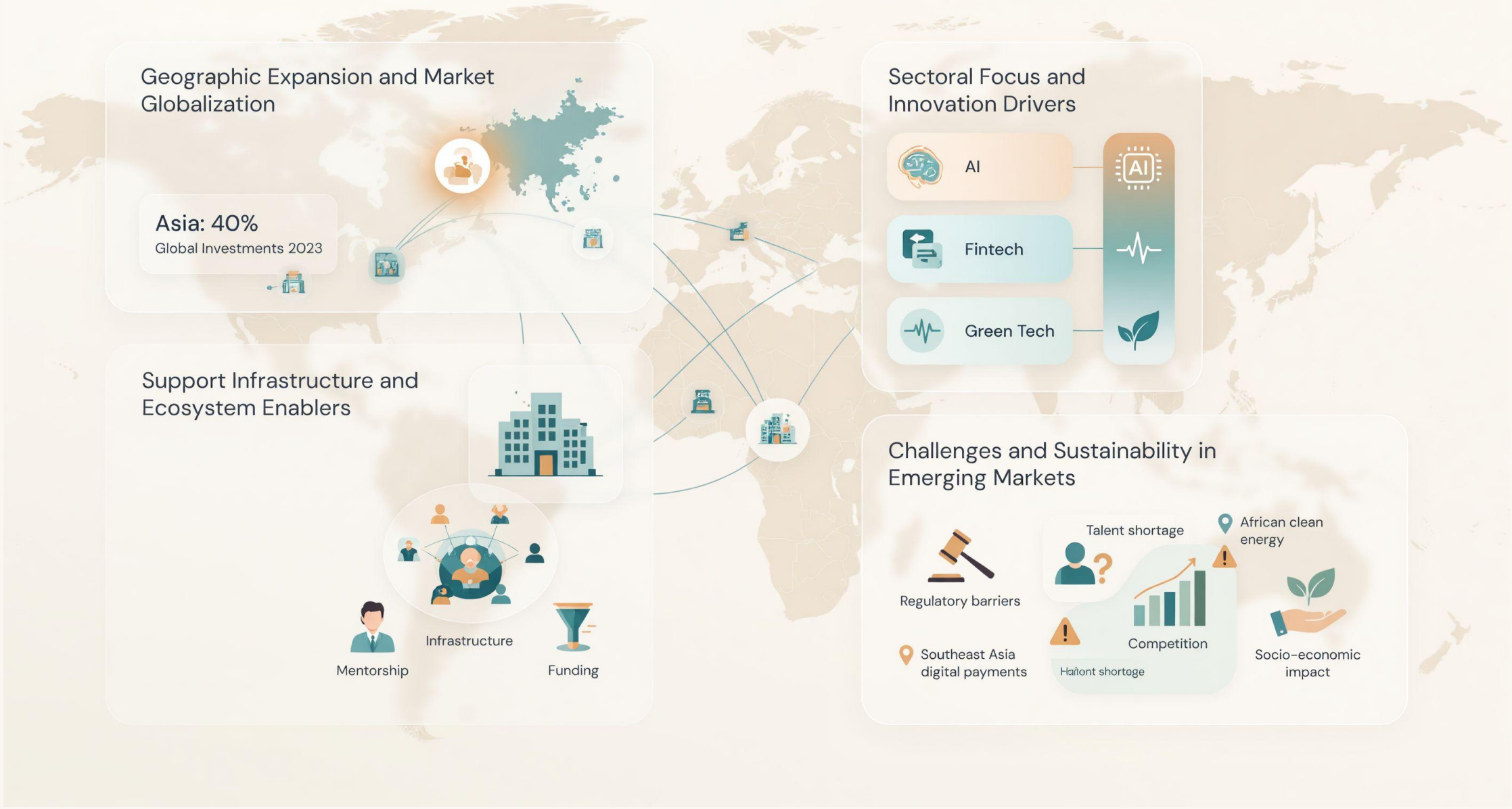
Chapter 4: Global Trend of Startup and Business

Ir Prof Alan Lam, JP

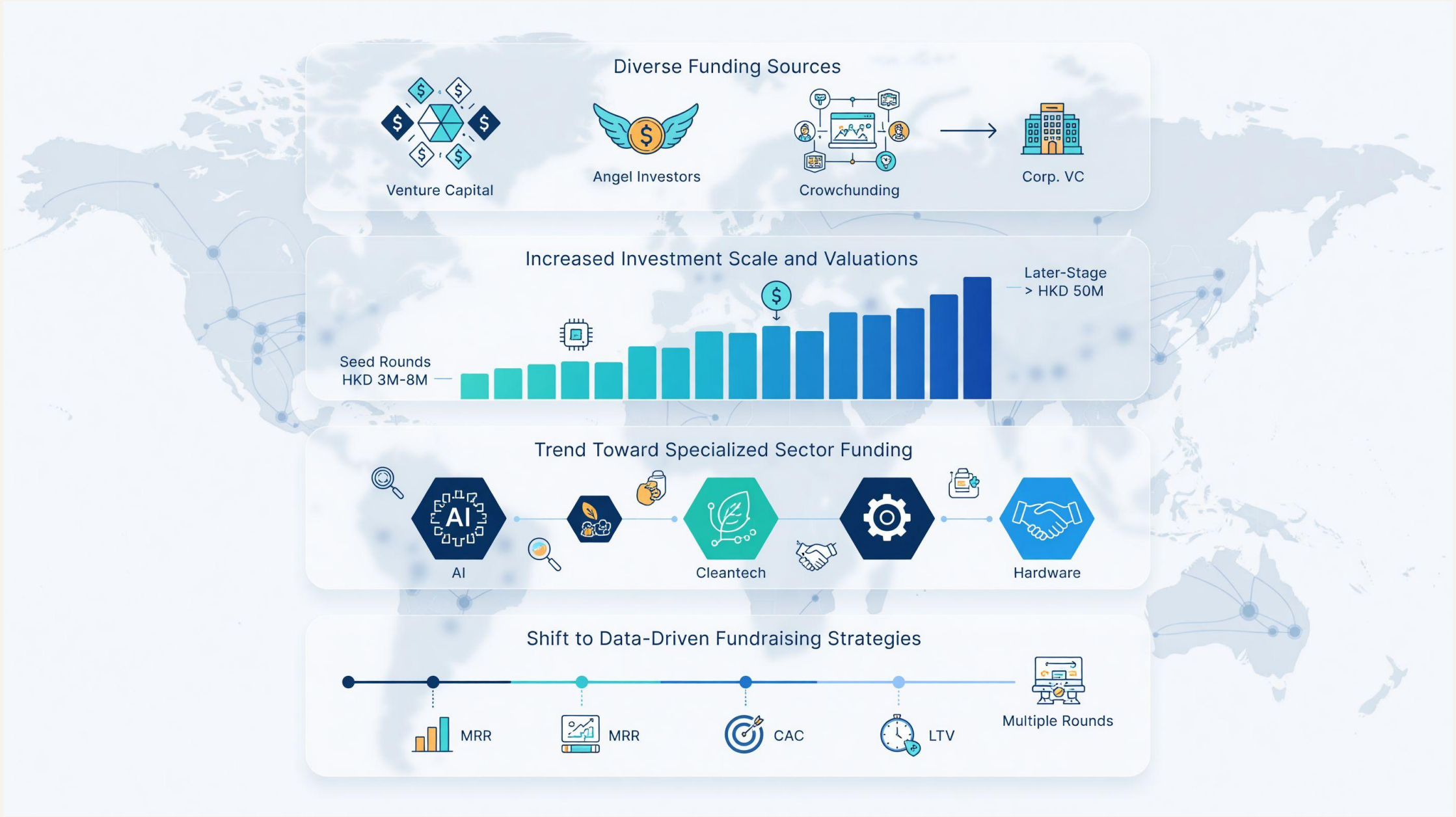


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Global Startup Landscape Analysis



Startup Financing Trends



Business Model Innovation in Startups



Impact of Technology on Business Models

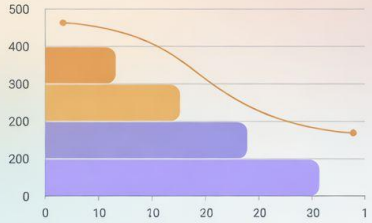
Digital Platforms & Cloud



AI & Data Analytics



Subscription Model with Churn



Monthly Subscribers <= 5580

Monthly Revenue HKD 1,116,000

New Revenue Streams & Innovation



Market Entry Strategies



Impact of Digital Transformation



Statistical Data on Startup Success Rates



Government Policies Supporting Startups



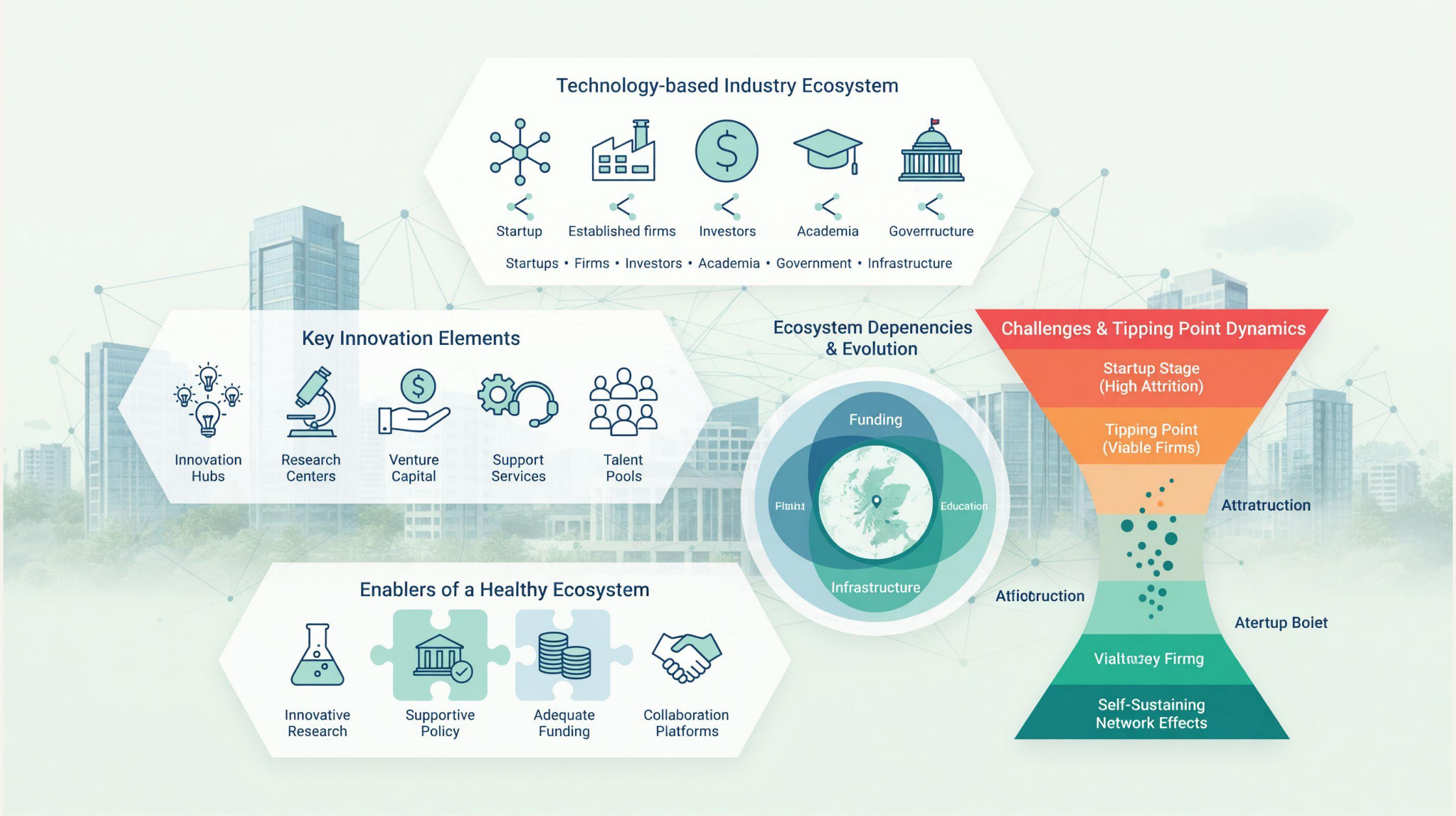
Chapter 5: Eco-system of Technology-based Industry

Ir Prof Alan Lam, JP



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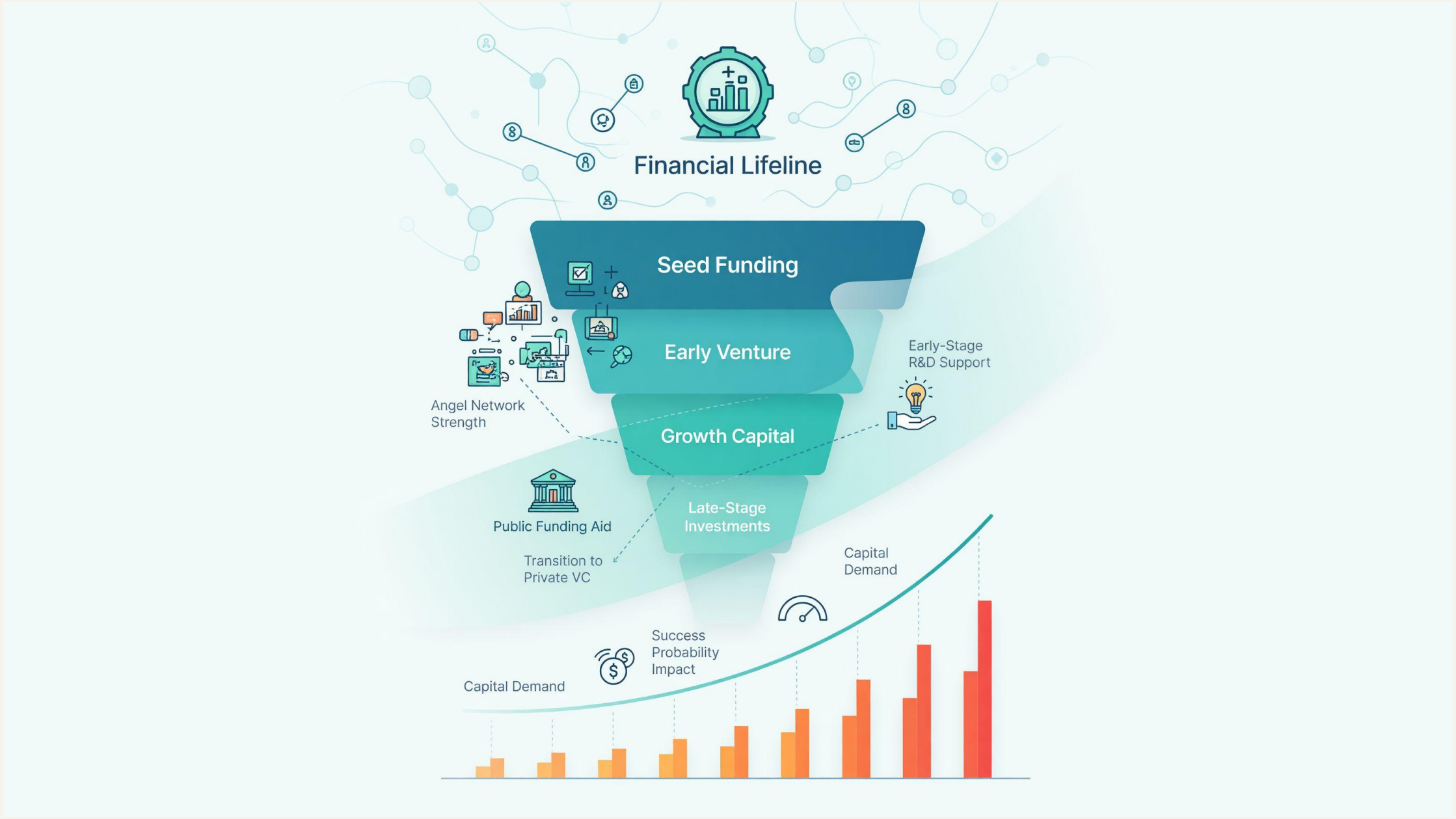
Definition and Components of Technology-based Industry Ecosystem



Academic and Research Contributions



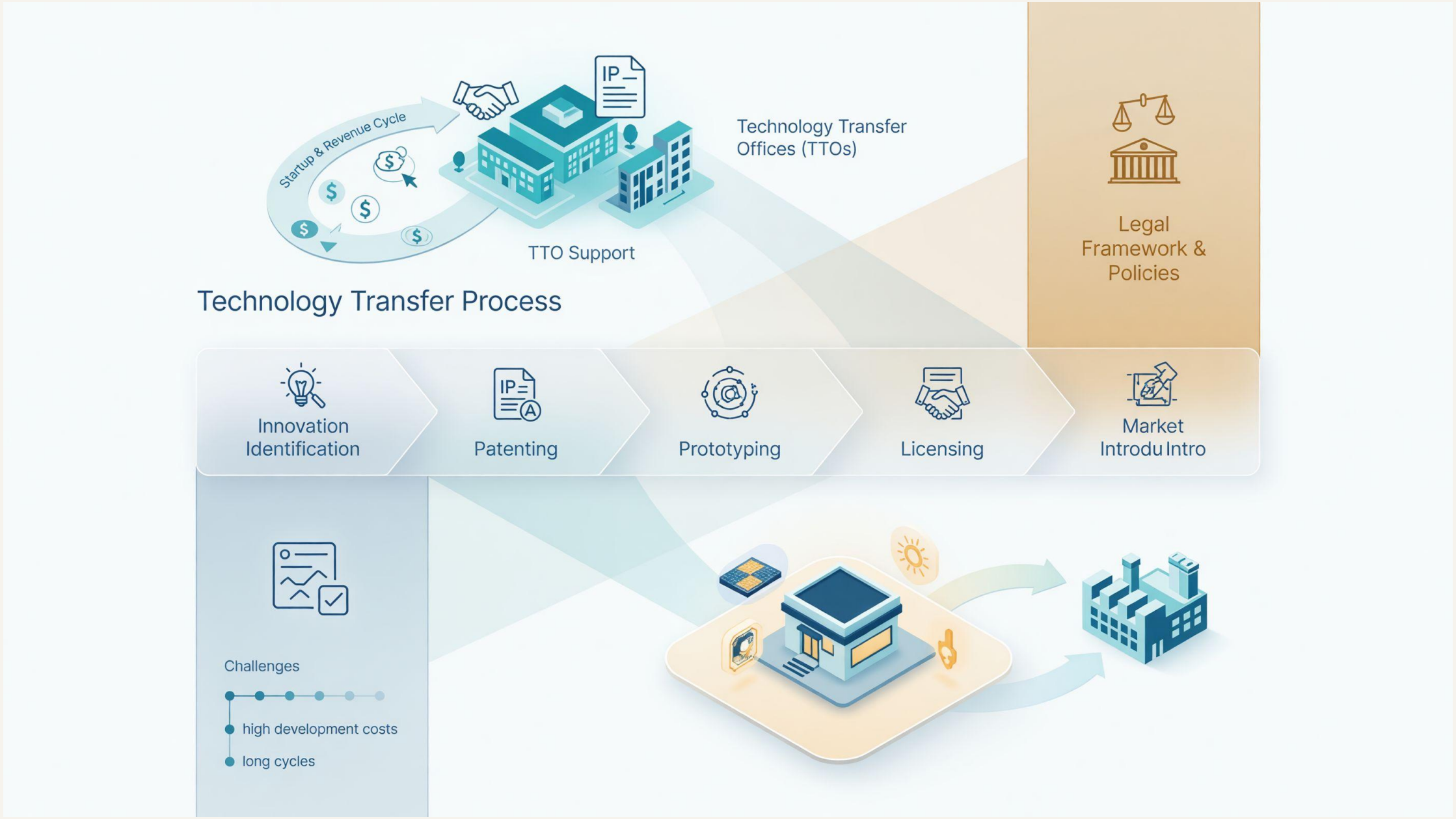
Venture Capital and Funding Networks



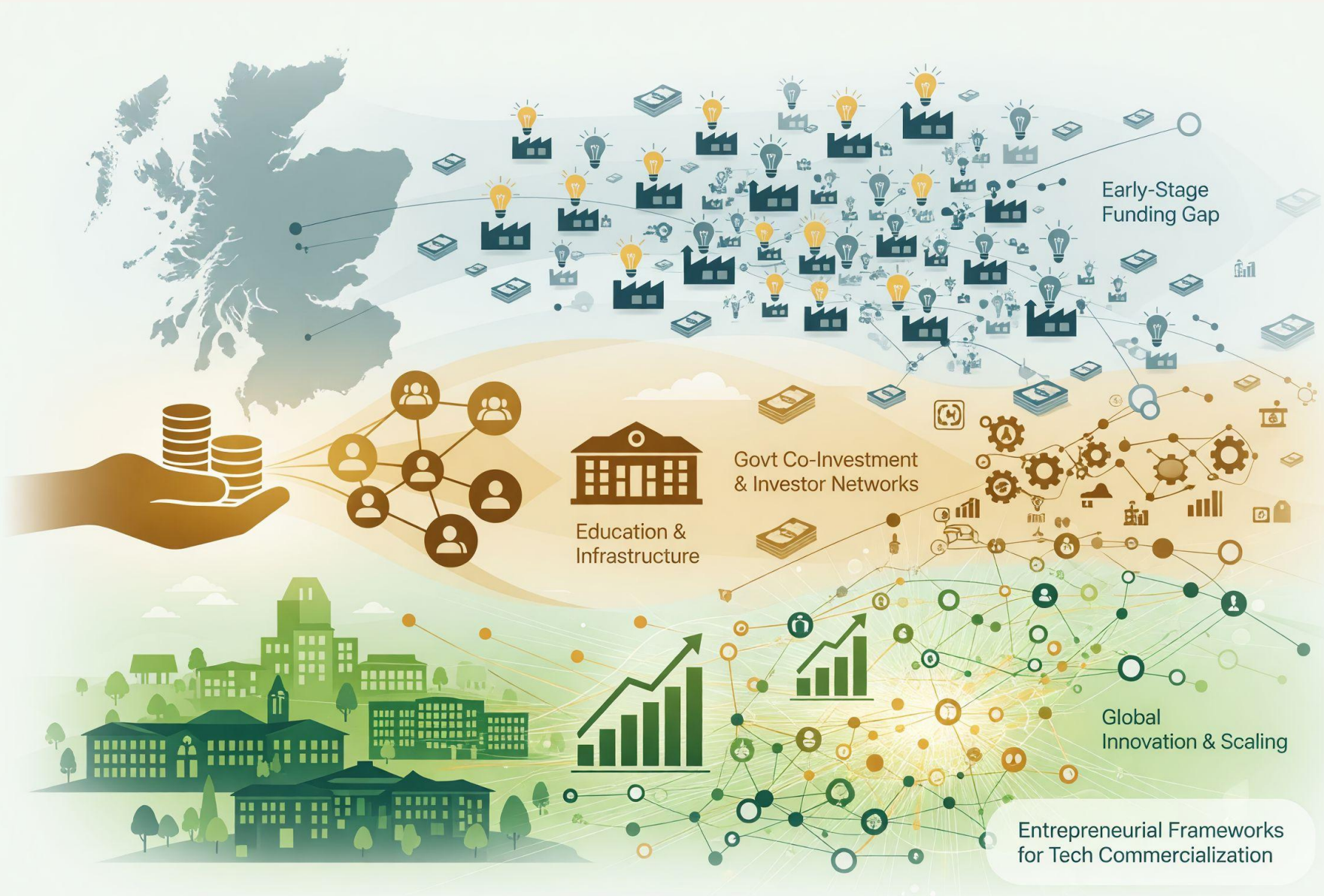
Collaborations and Partnerships



Technology Transfer and Commercialisation Processes



Case Studies on Technology Ecosystem Development



Chapter 6: Successful Example: DJI - Successful Drone Company

Ir Prof Alan Lam, JP



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Company Overview and History

DJI Corporate Evolution



- Founded 2006, Shenzhen
- Accessible & Reliable Drones
- 70-80% Global Market Share



Prank Wang

Technological Innovation Trajeentry



Flight Stabilization

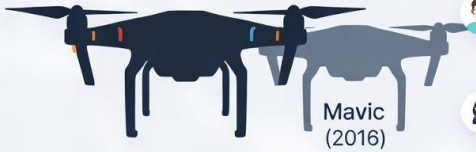


Battery Tech



Miniaturized Cameras

Product Milestones and Market Expansion



Mavic (2016)

Phantom (2013)



Flagship Series



Diverse Users



International Scaling

Strategic Challenges and Competitive Edge



Geopolitical Restrictions

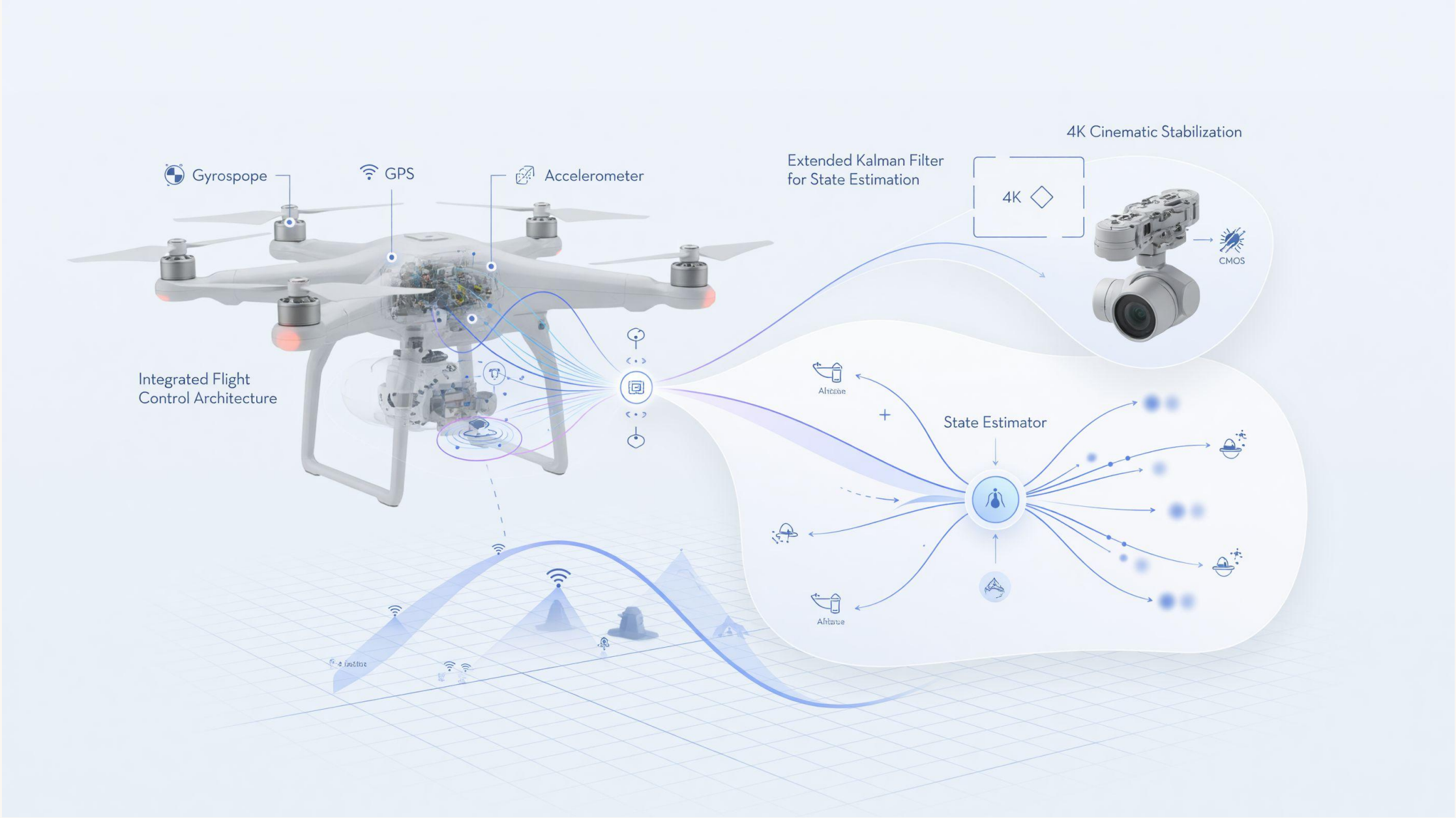


Persistent R&D

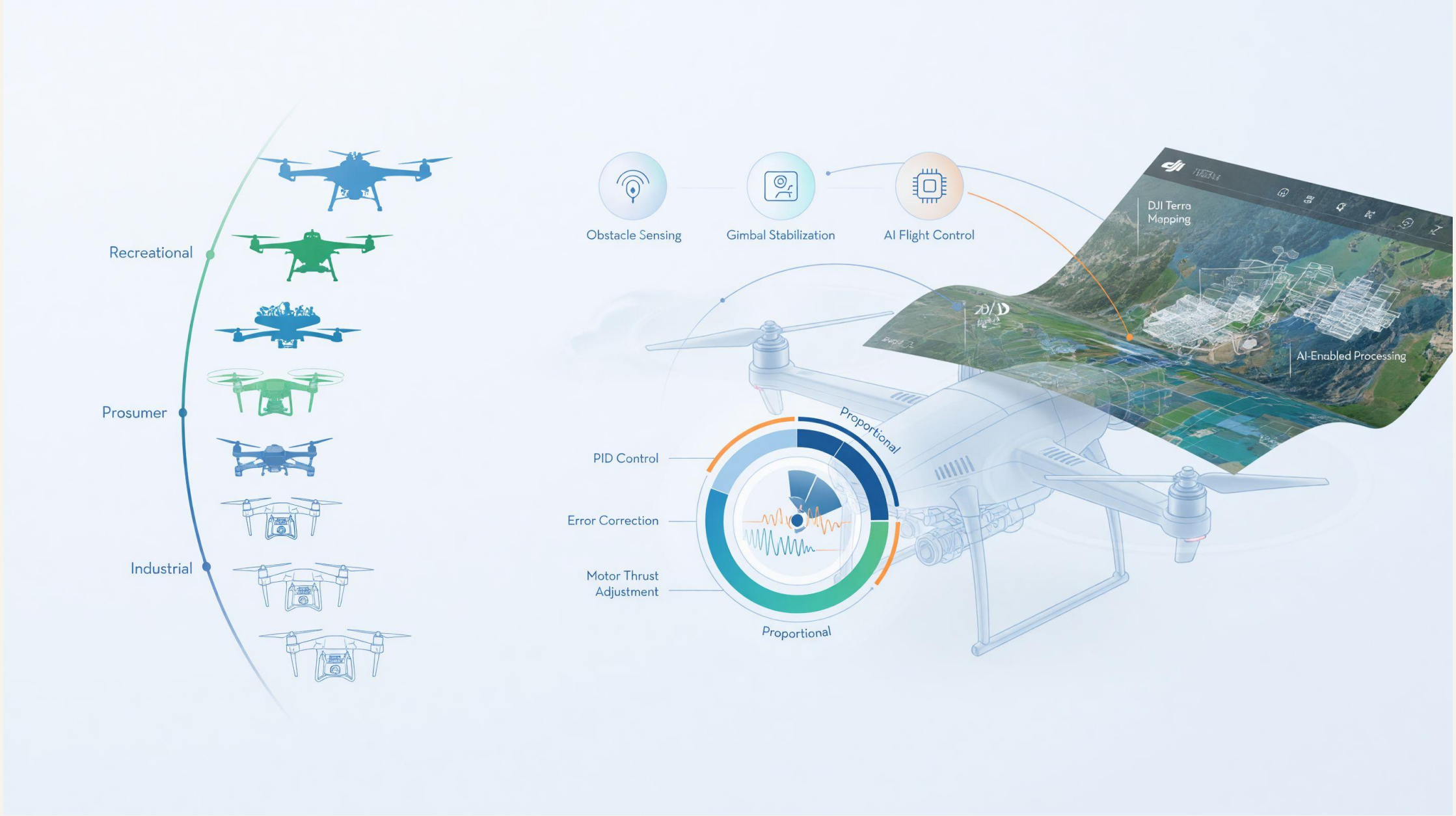


Market Leadership Strategy

Product Development and Innovation Pathway



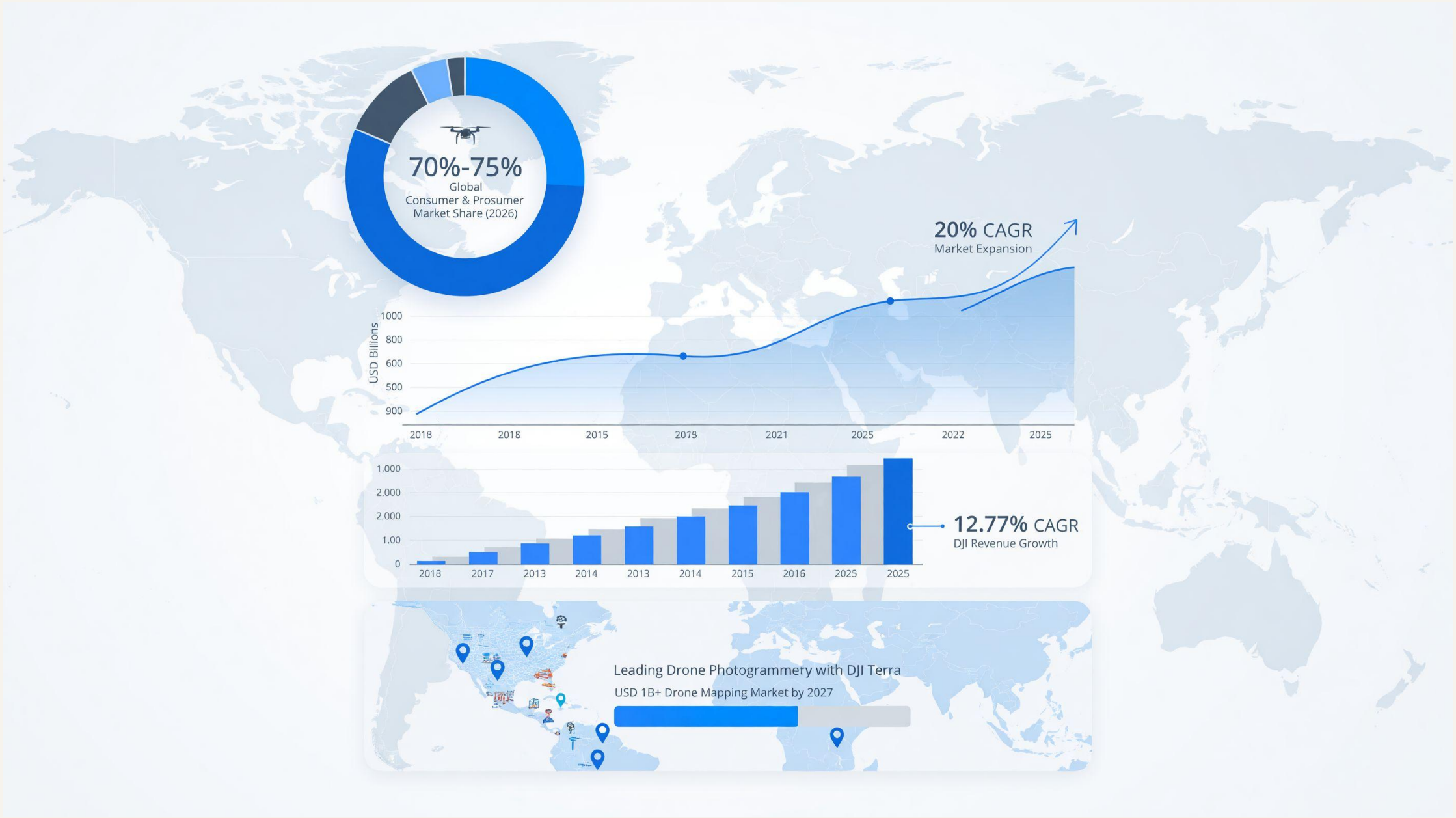
Product Portfolio and Innovation



Market Analysis and Strategy



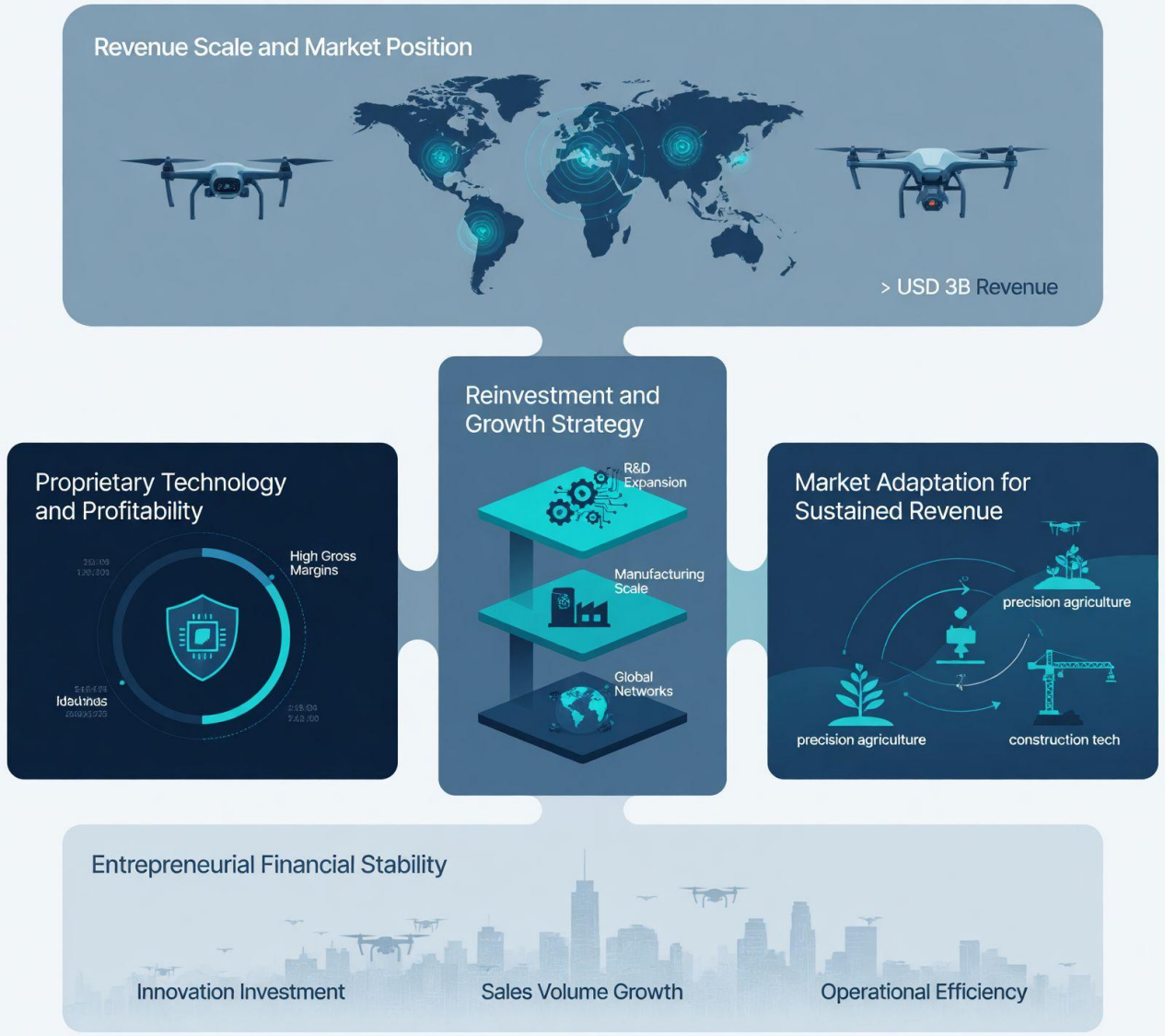
Quantitative Analysis of DJI's Market Share and Revenue Growth



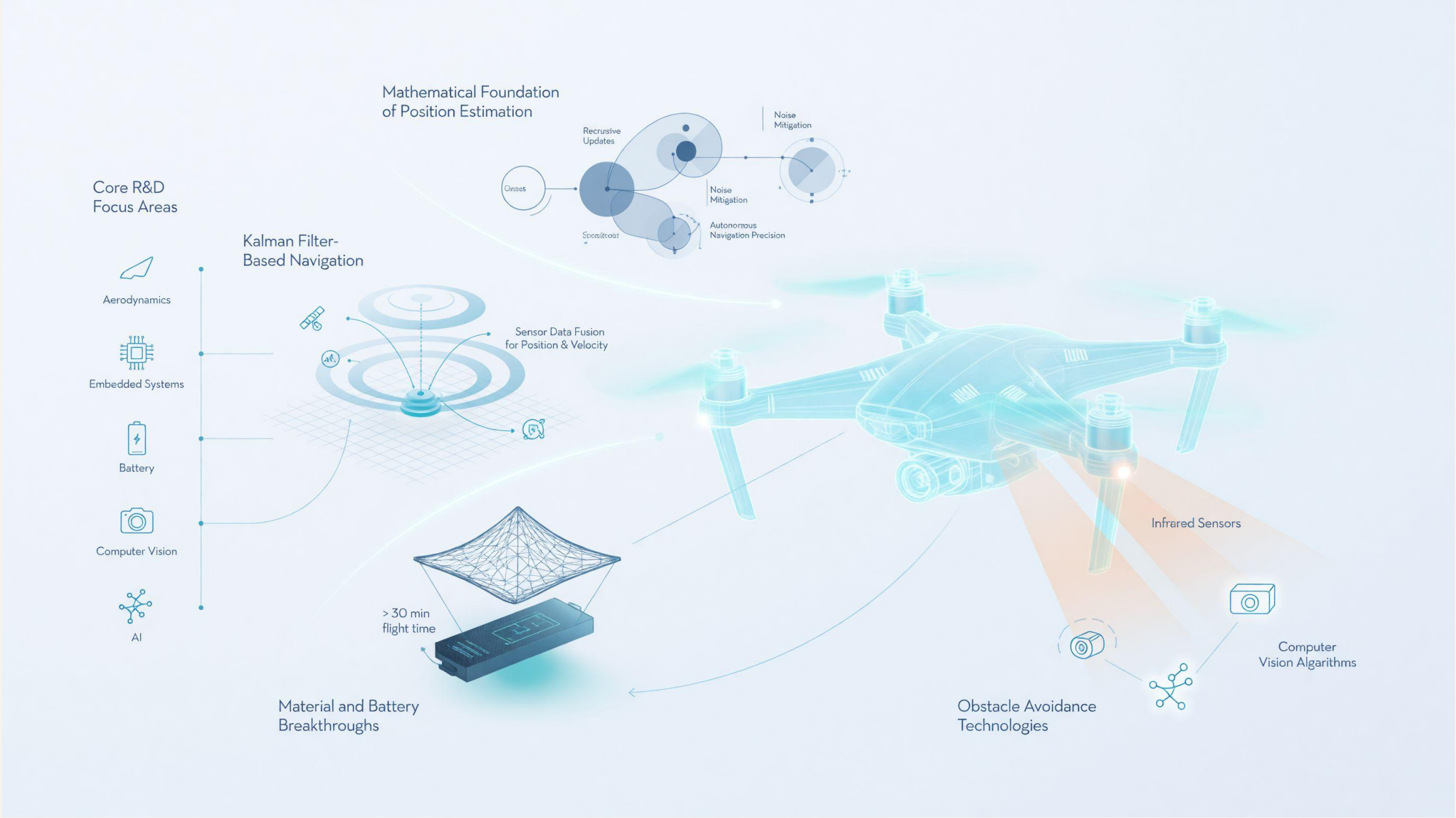
Business Model and Revenue Streams



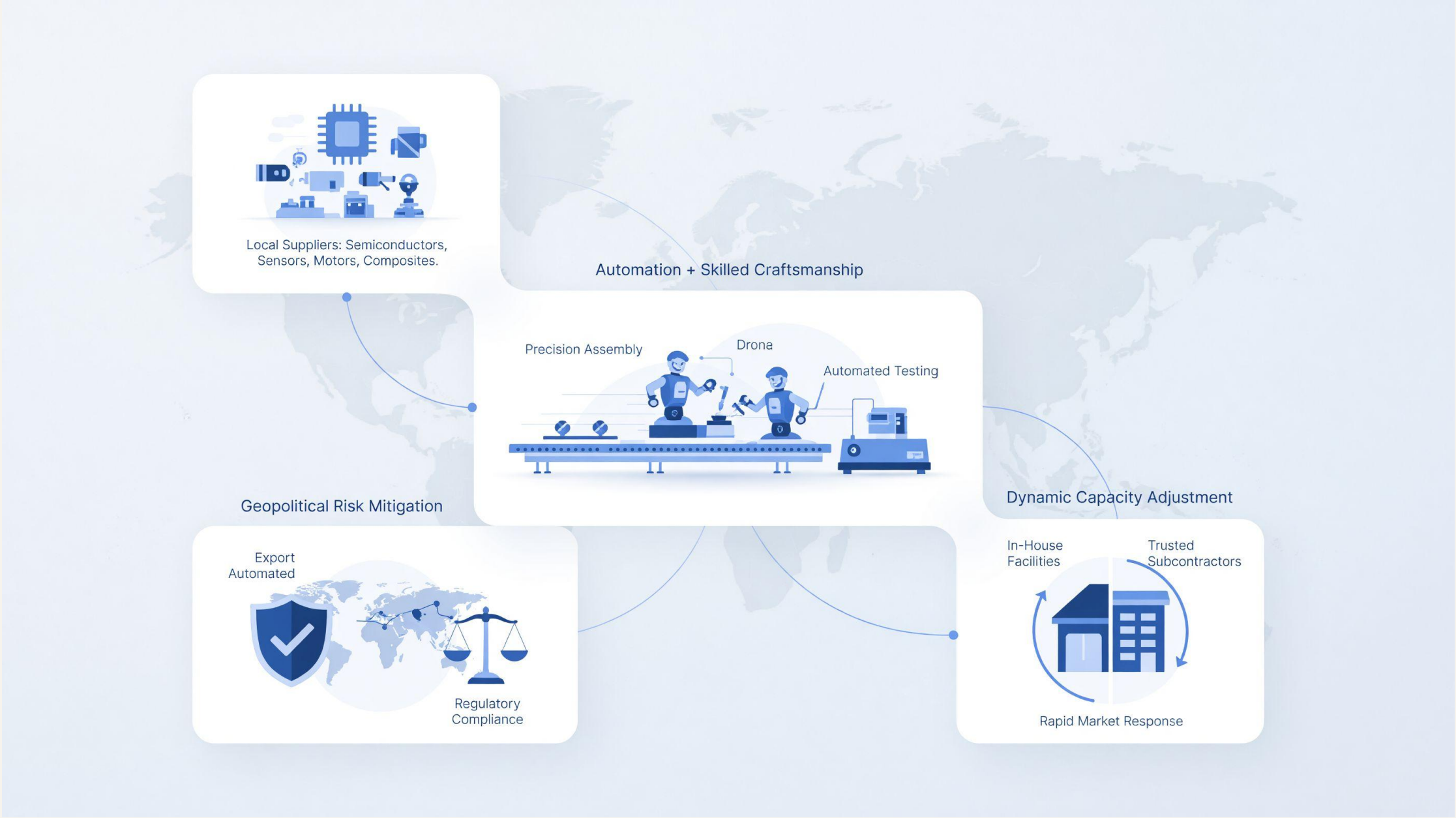
Financial Performance and Growth Metrics



Technological Breakthroughs and R&D



Supply Chain and Manufacturing Model



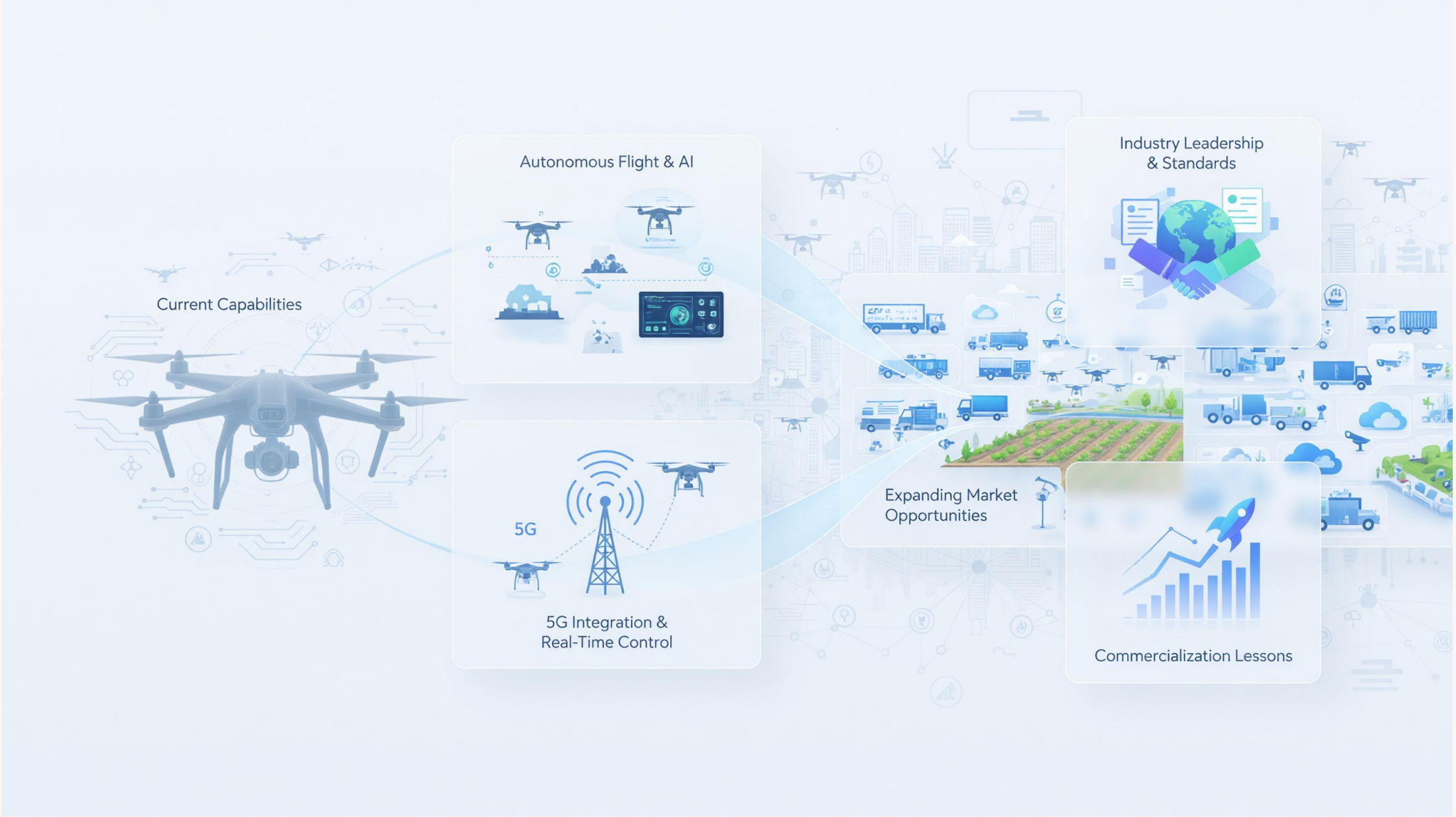
Global Expansion and Operations



Challenges and Risk Management



Future Prospects and Industry Impact



In-Class Discussion:

- Please based on the today's lecture to summarize what did you learn
- Based on your interest to think of some ideas about startup projects.
- Share your ideas.



Introduction to Entrepreneurship and Sustainable Business

Reference:

- The Secret To DJI's Market Leadership - DJI Viewpoints. https://viewpoints.dji.com/blog/the-secret-to-djis-market-leadership
- How U.S.-China Tariffs Could Reshape the Drone Industry (It's Not What You Think) - DRONELIFE. https://dronelife.com/2025/04/04/how-u-s-china-tariffs-could-reshape-the-drone-industry/
- Global Drone Market Report 2025-2030. https://droneii.com/product/drone-market-report?srsltid=AfmBOooe3YbfsRqZOuOQWqsH0gdE0ebaqVVyz1VAug_6FWBT2QBaIleX
- Top 10 Companies Leading the Drone Technology Market. https://blog.bccresearch.com/top-10-companies-leading-the-drone-technology-market



Ir Prof Alan Lam, JP
Adjunct Professor

End of Presentation
Thank You